Formalizing Mathematics and Making Its Women Visible

Valeria de Paiva, Topos Institute

For most of the twentieth century, formalized mathematics was easy to admire and easy to ignore. It belonged, many mathematicians felt, to a neighbouring world: logic, computer science, foundations. Proof assistants could certainly do impressive things, but they did not seem likely to alter everyday mathematical practice. Writing a paper, proving a theorem, checking a colleague's argument, teaching a graduate course—those were still human activities, carried out in ordinary mathematical language, with all the shortcuts and tacit understanding that such language permits.

That perception has begun to change. Formal mathematics—the representation of definitions, statements, and proofs in a language precise enough for a computer to verify every step—is no longer a curiosity at the edge of the discipline. It is becoming part of a broader conversation about mathematical reliability, digital infrastructure, and the role of artificial intelligence in research. If the past decade made proof assistants more usable, the current wave of AI has made them harder to dismiss. Machines can now generate mathematical prose, suggest proof strategies, search libraries, and in some settings produce formal solutions to Olympiad-style problems. The question is no longer simply whether formalization is possible. It is what formalization is for, how it changes mathematical practice, and why it may matter more in the age of AI than it ever did before.

What formalization is, and why it matters now

To formalize a theorem is to express it in a system such as HOL, Mizar, Rocq (formerly known as Coq), Agda, Isabelle, or Lean, together with enough definitions and proof steps for the system's kernel to check that the argument is correct. This notion of correctness is stronger, and narrower, than ordinary mathematical acceptance. It does not depend on whether a referee is convinced, whether a seminar audience nods along, or whether a gap seems "routine." A formal proof is one the machine can verify down to the axioms and inference rules. Of course, one still has to trust the software, but the point of proof assistants is that the trusted core can be made very small.

Formalization has a long prehistory. Whitehead and Russell's *Principia Mathematica* aimed at a hand-built version of the same dream: to reduce mathematics to explicit symbolic reasoning. Automated theorem proving and mechanized proof checking have been active research areas since the mid-twentieth century. Research-level mathematics has been formalized for decades, from the four-colour theorem to the Feit–Thompson theorem and, more recently, major projects in analysis, algebra, topology, and number theory. What is new is not the existence of proof assistants, but the sense that they may be moving from heroic demonstrations to practical infrastructure.

One reason is simply accumulation. Formal libraries are now large enough that formalization can become cumulative. Lean's Mathlib, developed by a worldwide community, contains an enormous body of undergraduate and research-level mathematics; Isabelle, Rocq, and other systems have built their own substantial ecosystems. Once definitions, standard lemmas, and common proof patterns are in place, the next theorem is cheaper to formalize than the last one. Another reason is improved tooling: better automation, better editors, more sophisticated tactics, better engineering (including e.g. blueprinting) and more experienced (zulip-organized) communities. The "de Bruijn constant"—the ratio between the size of a formal proof and the size of its informal counterpart—remains substantial, but it is less crushing than it once was. Some users report that the user experience has improved so dramatically in recent years that proof-writing now feels like a well-designed game, with the satisfying structure and immediate feedback that entails.

Paradox: why AI makes formalization more important

The third reason for progress is AI, and here the story is both exciting and genuinely paradoxical. There is now real progress in machine-assisted theorem proving and mathematical language generation. Large language models and related systems can propose tactics, guess intermediate lemmas, search vast formal libraries, help translate ordinary mathematical text into proof-assistant syntax, and even engage in real-time dialogue about mathematical problems. In professional mathematics, many researchers now find that conversational AI feels like working with a junior colleague—an extremely well-read one, available instantly, who can suggest approaches, clarify definitions, and help think through problems. For both theoretical and computational mathematicians, this kind of interaction is already changing how work gets done.

In 2024 DeepMind's AlphaProof achieved results on International Mathematical Olympiad problems that brought automated proving into the public eye, and the pace of development since then has been rapid. It would be a mistake to dismiss these advances as mere publicity.

It would be an equal mistake to overinterpret them. Solving Olympiad-style benchmark problems is not the same thing as understanding a research area, choosing the right definitions, organizing a long proof, or deciding which theorem is worth proving in the first place. Assessing the relevance of a mathematical contribution—deciding whether a result matters to a field, whether it opens new directions, whether it solves a problem people actually care about – has always been the hardest part of mathematical refereeing. No machine has shown the slightest competence at this. More importantly, the systems now grouped under the label "AI" are not reliable mathematicians. They are often brilliant assistants and terrible witnesses. They produce fluent explanations, plausible citations, and proof sketches that look persuasive even when they are wrong. They hallucinate. They conflate definitions. They silently smuggle in false steps. In ordinary mathematical life, human readers can sometimes patch these failures by intuition and background knowledge. But if machines begin generating mathematics at scale, that strategy will not be enough.

This is the paradox at the centre of the present moment. The better AI becomes at generating mathematical language, the more important formal verification becomes. Formalization is not made obsolete by machine-generated proofs; it becomes one of the only robust ways to tell whether a proof is genuine. If AI is generation, formalization is adjudication. If AI is a prolific source of conjectures, proof attempts, and formal snippets, proof assistants are what keep the resulting mathematical ecosystem from collapsing into a fog of persuasive but unverified claims.

Bridging formal and informal mathematics

There is a deeper question beneath all of this, one that has occupied my thinking for more than the past six years. Formal languages are powerful precisely because they are rigid: every symbol means exactly one thing, every rule applies in exactly one way. But this rigidity creates a profound cost. The flexibility of ordinary mathematical language—what we might call mathematical English, the lingua franca of the discipline—is not incidental to how we think. It is essential. We work with examples, intuition, partial sketches, and conversations in which meaning is negotiated rather than fixed. We use the same word in subtly different contexts. We rely on analogies and can shift our perspective when stuck. Mathematical English permits all of this.

Formal systems permit none of it. This gap is why formalization has remained laborious and why interoperability between different proof assistants, different libraries, and different formal systems is so difficult. Human mathematicians solve analogous communication problems by debating with each other—by negotiating meaning, trying different framings, building intuition together. Now, for the first time, machines are becoming quite good at this kind of seeming-to-reason-like-a-human: they can rephrase, they can offer multiple perspectives, they can spot when a term might mean different things to different communities.

My vision, which I still believe in, is to use the new tools that AI provides for language understanding to help bridge this gap. Not to replace formal systems—their precision is essential—but to help mathematicians and machines negotiate the passage between the rigidity of formal mathematics and the flexibility that mathematical communication requires. This is not about making formal mathematics less formal. It is about making the bridge between the two worlds less painful, less error-prone, and more natural. If AI can help with translation, with rephrasing, with making explicit the tacit steps that humans skip, then we may finally have the means to make formal mathematics a genuine part of ordinary mathematical practice rather than a specialized side-activity.

Surveying the landscape: the Hausdorff trimester

One of the clearest places I saw this shift taking shape was the Hausdorff trimester *Prospects of Formal Mathematics*, held at the Hausdorff Research Institute for Mathematics in Bonn in the summer of 2024. I co-organized the program with Kevin Buzzard, Jacques Carette, Michael Kohlhase, and Josef Urban. Before the trimester itself, over the years 2021–2024, the Topos Institute Colloquium in Berkeley had devoted a substantial series of talks to surveying the landscape of formalized mathematics and the varied approaches to it. The point was deliberate: to avoid taking for granted that any single system or approach defines the field. We invited speakers from Kevin Buzzard on Lean's mathematics library to Leonardo de Moura on Lean 4's design, from Josef Urban on machine learning over formal corpora to Georges Gonthier on organizing large proof libraries, from Jeremy Avigad's reflection on the state of the field to Gilles Dowek on proof-system interoperability and Lawrence Paulson on formalizing contemporary mathematics in Isabelle. That comprehensive conversation fed directly into the trimester itself.

For three months, Bonn became a meeting point for several communities that do not always occupy the same rooms: proof-assistant developers, category theorists, number theorists, homotopy type theorists, computer scientists, and researchers interested in machine learning for mathematics. Participants came from the worlds of Lean, Rocq, Isabelle, Agda, and beyond. Some were formalizing particular results; some were building libraries; some were trying to understand how large language models might help bridge between informal and formal mathematics. The commitment to genuine plurality—to not assuming that any one system had won or should win—was essential to what made the trimester work.

What made the trimester so striking was that it brought into focus the full ecology of formal mathematics. Formalization is not only about proving theorems in a proof assistant. It is also about designing and maintaining digital libraries of mathematics; making formal proofs readable enough that mathematicians can learn from them; finding ways to translate papers, lecture notes, and blackboard arguments into a form that machines can check; and asking what counts as a mathematical contribution when part of the labour has moved into code, library design, and verification. Those are questions about culture and institutions as much as about logic and software.

The trimester included a school for newcomers and workshops on formalization, on libraries of digital mathematics, and on the difficult relation between informal and formal proof. A recurring theme was that the real bottleneck may not be the logical core of proof assistants, but the interface between formal systems and ordinary mathematical life. If formal methods are to become mainstream, they must either become more hospitable to this way of working or be paired with tools that help translate between the two worlds. That is precisely where AI may be useful—not as a substitute for proof, but as a bridge.

Women in Formal Mathematics

Within that setting, together with Sandra Alves and EuroProofNet-Women, I organized a two-day workshop on **Women in Formal Mathematics**. This part of the story is as important as the technical one—and for deeper reasons than mere lack of representation. Formal mathematics sits at the intersection of mathematics and computer science, two disciplines that each already struggle with gender imbalance. The result is a field in which women can easily become invisible even while doing central work. And because formal mathematics is still young enough that its norms, prestige structures, and habits of collaboration are not yet fixed, visibility and intentional community-building matter enormously now.

The workshop was designed to make women's work in formal mathematics visible to the broader trimester community and to one another. We invited speakers whose work ranged across the landscape of the field: Ursula Martin on the social machinery of mathematics; Sandra Alves on type theory and lambda calculus; Brigitte Pientka on mechanized metatheory and dependently typed systems; María Inés de Frutos-Fernández on formalization in number theory; and Mateja Jamnik on trustworthy AI and automated reasoning. Contributed talks brought in younger researchers working on formal corpora, proof generalization, syntax and semantics, and the culture of formal proof. The aim was not simply to showcase a few successful careers, but to make visible a network of women participating in the construction of a field that is becoming increasingly important.

That visibility matters for intellectual reasons as well as social ones. Formal mathematics is not just a set of tools. It is a developing research culture with its own standards of elegance, its own forms of labour, and its own decisions about what counts as valuable work. Libraries do not build themselves. Interfaces do not design themselves. The translation between informal mathematical writing and formal verification does not happen automatically. If women are underrepresented in the rooms where those decisions are made, the field will inherit the same narrowness that many parts of mathematics and computing already suffer from. Conversely, if formal mathematics is being built now, this is exactly the moment when interventions in culture, mentorship, and visibility can have lasting effects.

What should mathematicians believe?

So what should mathematicians believe amid the current excitement? First, that something real is happening. Formalized mathematics is no longer confined to toy examples. There are now large libraries, serious theorem-proving environments, and major collaborations between proof-assistant communities and mainstream mathematicians. The experience of working with AI as a junior colleague in real time is changing mathematical practice, for better and worse, whether or not every theorem ends up fully formalized.

Second, that the loudest public claims should be treated with caution. We do not know what fraction of mathematics is formalized. We do not know how quickly research mathematics will move into proof assistants. We do know that formalization remains expensive, that interoperability is poor, and that benchmark success does not amount to autonomous mathematical understanding.

Third, and most importantly, we should recognize that AI makes the central question of formal mathematics sharper rather than fuzzier. Once machines can generate mathematical text at scale, mathematics needs reliable mechanisms for checking what those machines produce. And once mathematicians begin using AI as a thinking partner—as many already are—we need ways to distinguish genuine reasoning from plausible-sounding hallucination.

That is why I think formalization matters now in a way it did not even ten years ago. It is not merely a foundational exercise, or a hobby for logicians, or a badge of rigour attached to a few celebrated theorems. It is becoming part of the infrastructure by which mathematics may preserve trust in an era of machine generation. Proof assistants, formal libraries, and machine-checked verification will not replace mathematical creativity, taste, or explanation. But they may become essential supports for a discipline that increasingly has to distinguish insight from imitation, proof from persuasion, and genuine knowledge from very convincing statistical guesswork.

And if that is the future of formal mathematics, then the question of who gets to build it is not peripheral. Making women visible in formal mathematics is not only about fairness, though fairness would be reason enough. It is also about ensuring that one of the most consequential new interfaces between mathematics, computing, and AI is shaped by a community broad enough to deserve the name mathematical community at all.

Valeria de Paiva is a principal researcher at the Topos Institute in Berkeley, working in categorical logic, linear logic, and type theory. She created the Dialectica categories framework in her 1990 Cambridge PhD thesis, co-organized the Hausdorff trimester "Prospects of Formal Mathematics" and the Women in Formal Mathematics 2024 workshop, and is a founder and longtime organizer of the Women in Logic workshop series.